

Gopika Sree Kumar

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Professional Summary

M.S. in Data Science with 4 years of professional data experience, transitioning into AI/ML engineering through hands-on work building LLM-powered applications and machine learning systems. Experience with RAG over unstructured documents, retrieval optimization, LLM evaluation, NLP classification, and supervised ML, with strong foundations in Python, SQL, data engineering, and data platforms including Snowflake and BigQuery. Experienced in building and deploying ML applications using Flask/FastAPI and Docker.

Experience

Data Analytics & Research Intern — International Sibling Society (Remote, USA)

Mar 2026 – Present

- Contributed to an AI-powered SDG knowledge and classification workflow, combining document-based RAG over partnership MOUs, program reports, and SDG framework documents with LLM-based tagging of program records against the 17 SDGs.
- Worked on the retrieval pipeline, including document chunking, embeddings, retrieval, and prompt refinement to improve source-grounded answers; evaluated responses using a held-out set of 180 test questions and reviewed failure cases to identify areas for improvement.
- Built the supporting data workflow in BigQuery and dbt to refresh classifications and SDG coverage reporting, enabling staff to explore partnership activities and SDG representation across programs and countries.

Senior Business Analyst — Tiger Analytics

Apr 2022 – Jul 2024

- Partnered with a data science team at a banking client on an NLP classification solution for customer chat and call transcripts, contributing to text preprocessing, feature development, and analysis of category-specific language patterns.
- Cleaned and normalized transcript data using Python (pandas, regex, NLTK) and SQL, profiled category distributions, and developed keyword and phrase features used to distinguish Solved, Unresolved, and Deferred interactions.
- Reviewed training and validation examples with the data science team and analyzed misclassified transcripts to identify recurring failure patterns, using those findings to refine features and keyword rules across model iterations.
- Engineered an automated ETL pipeline using Python, SQL, and Snowflake, with stored procedures generating 100+ tables and scheduled daily refreshes, eliminating 40+ hours/month of manual work.
- Designed relational and dimensional data models and KPI frameworks from complex business and security requirements, supporting analytical and operational reporting.
- Delivered compliance and operational dashboards using Tableau and Sigma, reducing incident investigation time by 60% through improved visibility into security and operational data.
- Collaborated with data science, engineering, and business teams in Agile delivery cycles to develop and deliver production-ready analytical datasets.

Assistant System Engineer — Tata Consultancy Services

Nov 2020 – Mar 2022

- Supported a large-scale Teradata-to-Snowflake migration for a UK banking client, performing SQL conversion, data validation, reconciliation, and testing across migrated datasets.
- Investigated migration issues involving date conversions, joins, dependent objects, and schema differences, working with QA and data-governance teams to resolve discrepancies and maintain data integrity.
- Improved query performance by approximately 25% through SQL optimization and validation of migrated workloads.

Projects

Warehouse Copilot — Text-to-SQL Agent over Snowflake Python · LLM · RAG · Vector DB · Snowflake

[GitHub](#)

- Built an agentic text-to-SQL system that translates natural-language questions into SQL against a Snowflake dimensional warehouse, using retrieved schema context to guide query generation.
- Implemented embedding-based schema retrieval over table and column descriptions so the LLM receives only relevant database context, with read-only guardrails and query-cost limits to prevent destructive or runaway queries.
- Added a self-correction loop that analyzes SQL execution errors and iteratively repairs generated queries before returning results.
- Built an evaluation framework measuring SQL execution accuracy, answer quality, latency, and inference cost across a question-to-SQL benchmark.

Large-Scale Demand Forecasting with XGBoost *Python · XGBoost · Flask · Docker*

[GitHub](#)

- Built an end-to-end demand forecasting system using the M5 Walmart dataset, covering 3,049 products and approximately 58M item-day records; XGBoost achieved a test RMSE of 1.92 on a chronologically held-out 28-day window.
- Engineered leak-free time-series features including lag, rolling statistics, price, calendar/event, and SNAP features, using shift-before-rolling transformations to prevent target leakage.
- Tuned the model using RandomizedSearchCV, TimeSeriesSplit, and early stopping, reaching a 26% improvement in RMSE over a naive forecasting baseline.
- Deployed the forecasting model as a Flask REST API containerized with Docker, loading saved encoders and feature medians to reduce train/serve skew.

Mahabharata Intelligence — Graph RAG over Epic Text *Python · LLM · Knowledge Graph · FAISS*

[GitHub](#)

- Built a Graph RAG system over the unabridged Mahabharata (2,328 pages, 2.55M words), combining a knowledge graph of 378 entities and 2,805 relations with FAISS vector search over 9,280 page-cited chunks.
- Developed the graph pipeline with regex and LLM-based relationship extraction, LLM-verified alias resolution, and a query planner that classifies questions into 8 types and links epithets to canonical entities for multi-hop traversal.
- Implemented a hybrid reranker scoring semantic similarity, graph score, entity overlap, and relation match; answers cite evidence per claim, with citation validation, confidence downgrades for hedged or uncited claims, and abstention.
- Ran an LLM-judged ablation of vector, graph, and hybrid retrieval: hybrid reached 0.70 relation recall and 1.00 path accuracy versus 0.00 for vector-only, with 1.00 citation validity; backed by 274 pytest tests.

Technical Skills

Generative AI & LLMs: LLM APIs (OpenAI, Anthropic), RAG, Embeddings, Vector Databases (FAISS, Chroma), AI Agents & Tool Use, Prompt Engineering, Structured Outputs, LLM Evaluation, Guardrails

Machine Learning & NLP: XGBoost, scikit-learn, Logistic Regression, NLP, NLTK, Text Preprocessing, Feature Engineering, Hyperparameter Tuning, Classification, Regression, Time-Series Modeling, Model Evaluation, ROC-AUC, PR-AUC, RMSE

ML Engineering: Python, pandas, NumPy, SciPy, Flask, FastAPI, Docker, REST APIs, Pydantic, Git, Model Deployment, Inference Optimization

Data Engineering & Cloud: SQL, Snowflake, BigQuery, dbt Core, Kafka, Spark Structured Streaming, MongoDB, ETL/ELT, Dimensional Modeling, Data Quality, GCP, AWS

Analytics & Statistics: Tableau, Sigma, Hypothesis Testing, A/B Testing, Statistical Analysis

Education

University at Buffalo, SUNY — Buffalo, NY

Aug 2024 – Feb 2026

M.S. in Data Science

BNM Institute of Technology (BNMIT) — India

Aug 2016 – Oct 2020

B.E. in Electronics & Communication Engineering